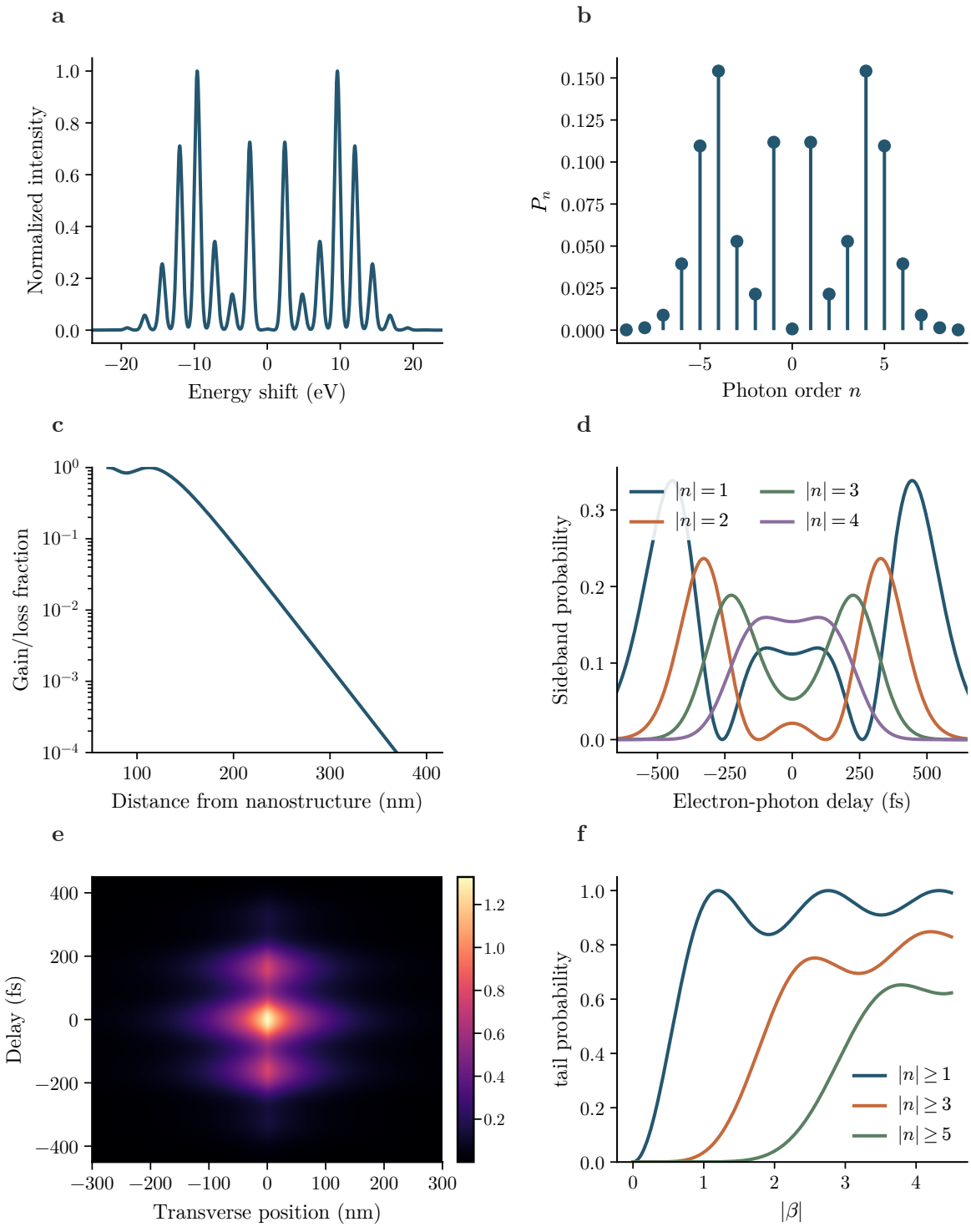

a
Normalized intensity
1.0
0.8
0.6
0.4
0.2
0.0
−20
−10
0
10
20
Energy shift (eV)
b
P_n
0.150
0.125
0.100
0.075
0.050
0.025
0.000
−5
0
5
Photon order n
c
Gain/loss fraction
10^0
10^{-1}
10^{-2}
10^{-3}
10^{-4}
100
200
300
400
Distance from nanostructure (nm)
d
Sideband probability
$|n| = 1$
$|n| = 2$
$|n| = 3$
$|n| = 4$
0.3
0.2
0.1
0.0
−500
−250
0
250
500
Electron-photon delay (fs)
e
Delay (fs)
400
200
0
−200
−400
−300
−200
−100
0
100
200
300
Transverse position (nm)
1.2
1.0
0.8
0.6
0.4
0.2
f
tail probability
1.0
0.8
0.6
0.4
0.2
0.0
$|n| \geq 1$
$|n| \geq 3$
$|n| \geq 5$
0
1
2
3
4
$|\beta|$